a Single-cell model development

Reference cohort

GSE178341

64 tumor samples

Reference cell annotations

Subtypes: at least 50 cells

Feature construction

Cell ecology

71 → 24 → top 18 subtype proportions

CNV-candidate epithelial MLH1

Mean expression

CNV-candidate / epithelial weighting

scRNA classifier

Z-score standardization

Shrinkage LDA

18 subtype proportions

+ CNV-candidate weighted MLH1

Model assessment

Patient-grouped five-fold

Fixed 18-subtype panel

TOSICA cell-type annotation

GSE132465 + GSE200997

Retrospective evaluation

Reuse the cell-feature definitions

Single-cell feature transfer

b Bulk model development

Bulk expression

TCGA-COADREAD + GSE39582

BayesPrism

12 subtype features from 17 candidates

Cohort-held-out MMR4 + fixed fusion

MLH1 MSH2 MSH6 PMS2

MLP (4 units)

fixed 0.5:0.5 fusion

MMR4: fit on 7; evaluate held-out cohort

MMR4 held out by cohort; subtype scores fixed

c Exploratory spatial transfer

Spatial counts

VISDS

14 sections / 7 patients

cell2location

18 fractions + weighted MLH1 proxy

MSI/dMMR inference

18 subtypes + MLH1

Patient-level scores

Reference-based reconstruction → patient aggregation

Technical transfer · 7 patients / 2 positives

Schematic glyphs; external evaluation is retrospective. Spatial groups are inferred, and measurement equivalence is unestablished.